

0 4

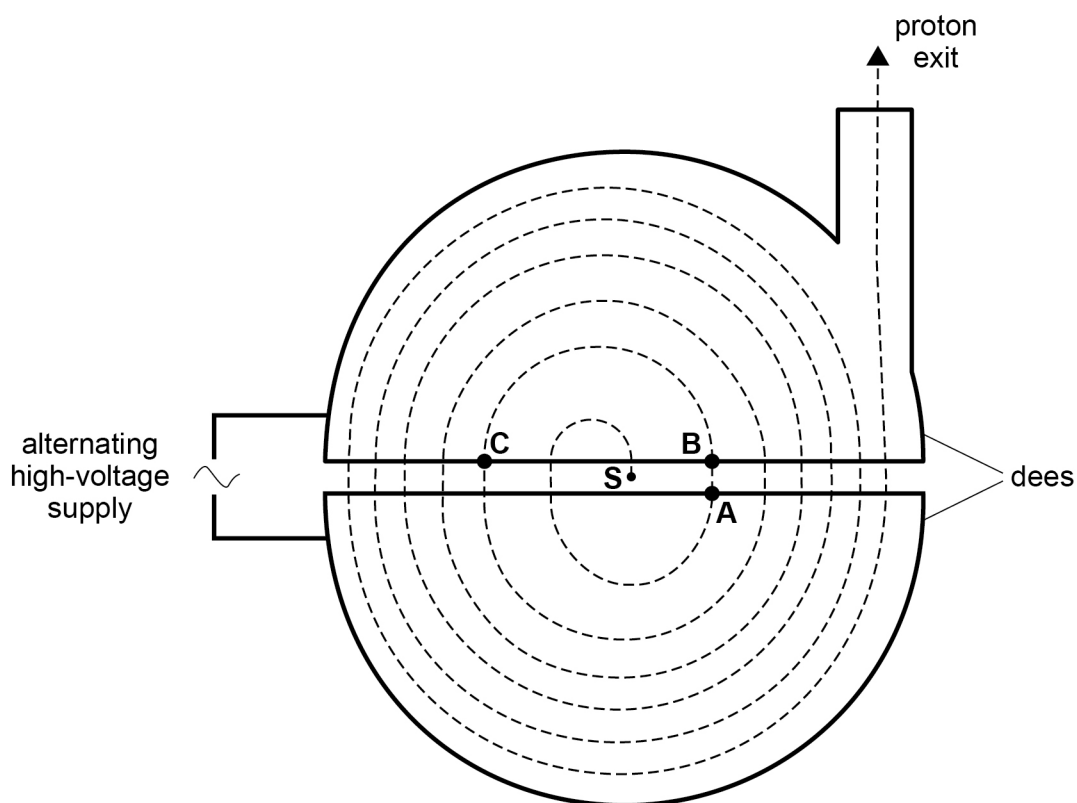
**Figure 7** is a plan view of a cyclotron. Two hollow semi-circular electrodes, called dees, lie within a uniform magnetic field. An alternating high-voltage supply is connected to the dees.

A proton enters the cyclotron at **S** and moves in the plane of the paper. The dashed line shows the path of the proton.

Three positions, **A**, **B** and **C**, along the path are marked.

- The proton crosses a potential difference between **A** and **B**.
- The proton moves within the uniform magnetic field between **B** and **C**.

**Figure 7**



0 4

1

State the direction of the magnetic field in **Figure 7**.

[1 mark]

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**0 4 . 2** Explain what happens to the proton as it crosses from **A** to **B**.

**[2 marks]**

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**0 4 . 3** Explain the path of the proton as it moves from **B** to **C**.

**[3 marks]**

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**Question 4 continues on the next page**

**Turn over ►**



**0 4 . 4** The angular speed  $\omega$  of the proton in the cyclotron is

$$\omega = \frac{Bq}{m}$$

where  $B$  is the magnetic flux density in the cyclotron  
 $q$  is the charge of the proton  
 $m$  is the mass of the proton.

Show that the kinetic energy  $E_k$  of the proton as it exits the cyclotron is

$$E_k = \frac{(BqR)^2}{2m}$$

where  $R$  is the maximum radius of the proton path in the cyclotron.

**[2 marks]**



0 4 . 5

In one cyclotron, protons exit with  $E_k = 10 \text{ MeV}$  when  $R = 0.34 \text{ m}$ .

Calculate the magnetic flux density in the cyclotron.  
Ignore relativistic effects in your calculation.

[3 marks]

magnetic flux density = \_\_\_\_\_ T

11

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Turn over ►



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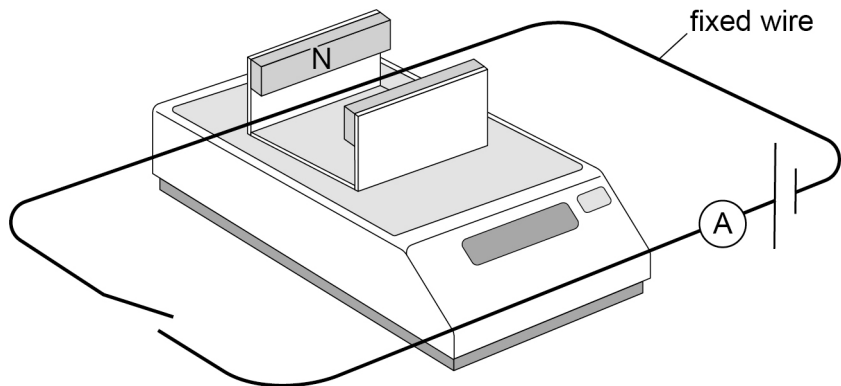
2 5

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The diagram shows equipment used to investigate the force produced by a current in a wire that is in a uniform magnetic field. The wire is fixed and cannot move.

Each magnet is 5 cm long and the magnetic flux density of the field between the magnets is 120 mT.

The north pole of one magnet is labelled in the diagram.



The mass balance reads zero when there is no current in the wire.

What is the reading, in gram, on the mass balance when the current is 2.0 A?

[1 mark]

**A** -1.2

**B** -0.12

**C** +0.12

**D** +1.2

**2****6**

What is the fundamental (base) SI unit for magnetic flux density?

[1 mark]

**A**  $\text{kg A}^{-1} \text{s}^{-2}$

☐

**B**  $\text{kg A}^{-1} \text{s}^2$

☐

**C**  $\text{kg C}^{-1} \text{s}^{-3}$

☐

**D**  $\text{kg m A}^{-1} \text{s}^{-2}$

☐

Turn over ►



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A proton and an alpha particle have the same kinetic energy.

The proton and the alpha particle enter a uniform magnetic field at  $90^\circ$  to the magnetic field lines. Both particles experience a force due to their motion in the magnetic field.

What is  $\frac{\text{the magnitude of the force on the proton}}{\text{the magnitude of the force on the alpha particle}}$ ?

[1 mark]

**A** 0.25

☐

**B** 0.50

☐

**C** 1

☐

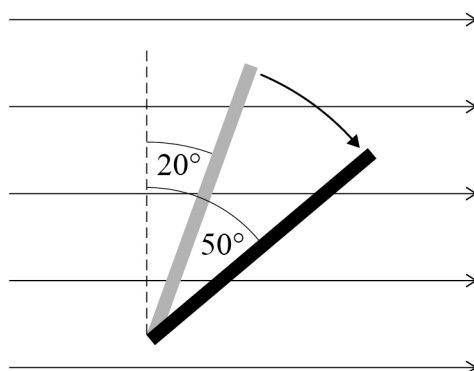
**D** 2

☐

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The diagram shows the end view of a rectangular coil in a uniform magnetic field of flux density 70 mT.

The plane of the coil makes an angle of  $20^\circ$  to the magnetic field lines.  
The coil has 50 turns and an area of  $0.035 \text{ m}^2$ .



The coil is rotated so that the plane of the coil makes an angle of  $50^\circ$  to the magnetic field lines.

As a result of this change, the magnetic flux linkage

[1 mark]

**A** decreases by  $3.6 \times 10^{-2} \text{ Wb}$ .

☐

**B** decreases by  $1.1 \times 10^{-1} \text{ Wb}$ .

☐

**C** increases by  $3.6 \times 10^{-2} \text{ Wb}$ .

☐

**D** increases by  $1.1 \times 10^{-1} \text{ Wb}$ .

☐

Turn over for the next question

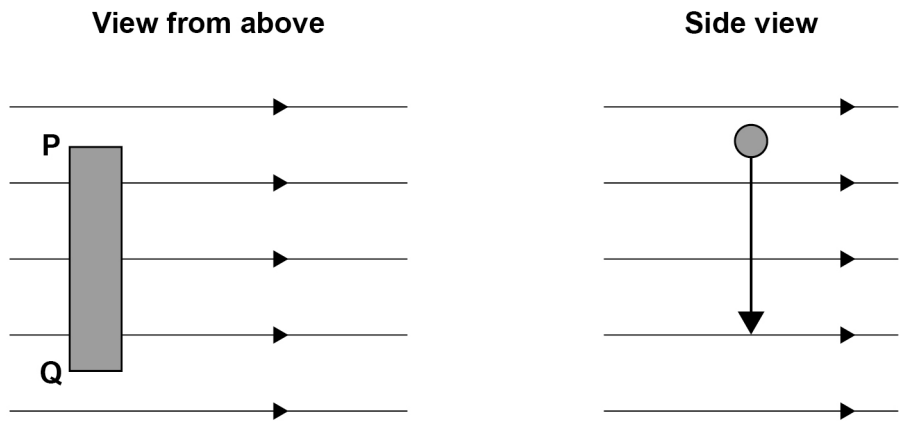


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3 0

A metal rod falls at a constant velocity perpendicular to a uniform magnetic field. An emf is induced between the ends of the rod, **P** and **Q**.

The rod is 20 cm long and travels 60 cm through the field in 0.5 s.  
The magnetic flux density is 400 mT.



Which row shows the emf induced across the ends of the rod and the polarity of end **P** and end **Q**?

[1 mark]

|          | emf / mV | Polarity of P | Polarity of Q |                          |
|----------|----------|---------------|---------------|--------------------------|
| <b>A</b> | 48       | Positive      | Negative      | <input type="checkbox"/> |
| <b>B</b> | 96       | Positive      | Negative      | <input type="checkbox"/> |
| <b>C</b> | 48       | Negative      | Positive      | <input type="checkbox"/> |
| <b>D</b> | 96       | Negative      | Positive      | <input type="checkbox"/> |



**Turn over for the next question**

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outside the  
box*

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ANSWER IN THE SPACES PROVIDED**

**Turn over ►**



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Which is a scalar quantity?

[1 mark]

**A** change in momentum

☐

**B** absolute electric potential

☐

**C** gravitational field strength

☐

**D** magnetic flux density

☐

**Turn over for the next question**

**Turn over ►**

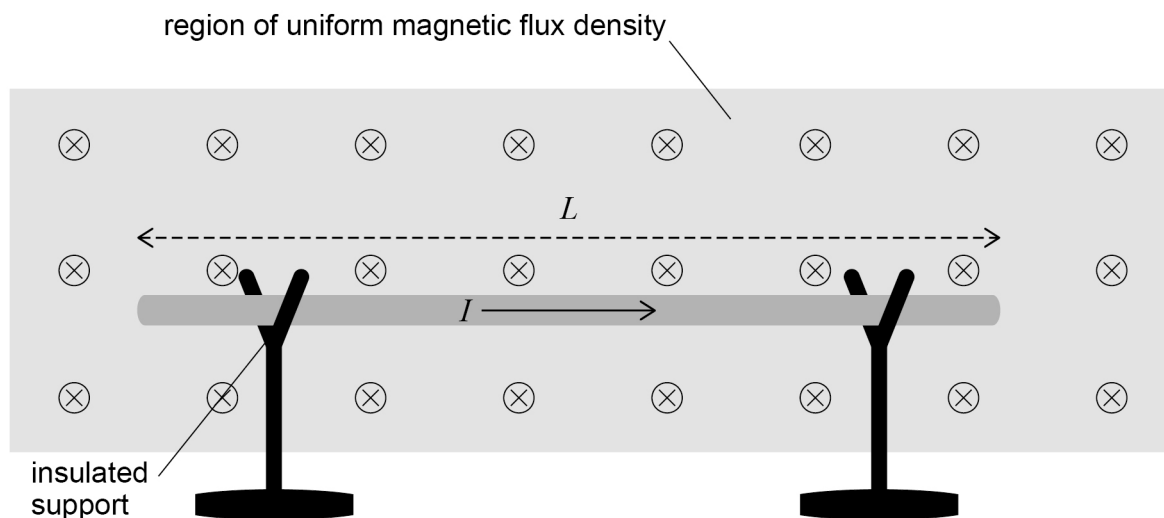


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A uniform wire of length  $L$  rests on two insulated supports within a uniform magnetic field.

The magnetic flux density  $B$  acts horizontally and perpendicularly to the wire.

The cross-sectional area of the wire is  $A$  and the density of the wire is  $\rho$ .



The electrical connections to the wire are not shown.

The wire just lifts off the supports when there is a current  $I$  in the wire.

What is the magnitude of  $I$ ?

[1 mark]

A  $\frac{\rho g}{BA}$  ☐

B  $\frac{\rho g A}{B}$  ☐

C  $\frac{\rho g A}{BL}$  ☐

D  $\frac{\rho g AL}{B}$  ☐



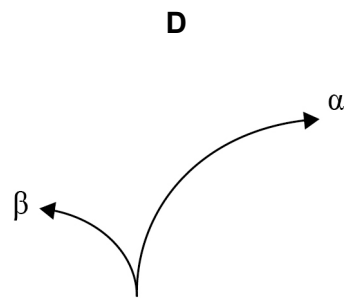
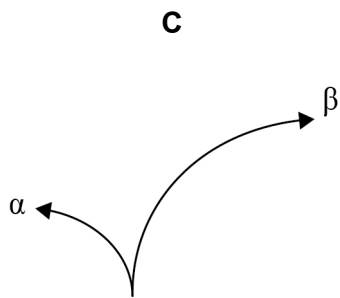
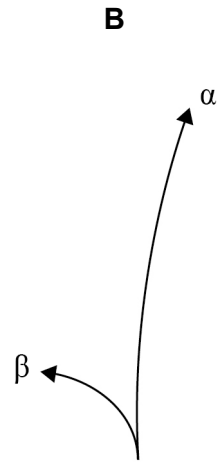
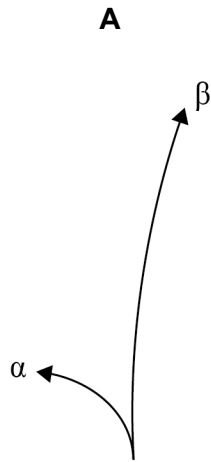
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An alpha particle and a beta particle separately enter a uniform magnetic field at  $90^\circ$  to the field lines.

Both particles enter the field with the same velocity.

Which diagram best shows the paths of the particles in the magnetic field?

[1 mark]



**A** ☐

**B** ☐

**C** ☐

**D** ☐

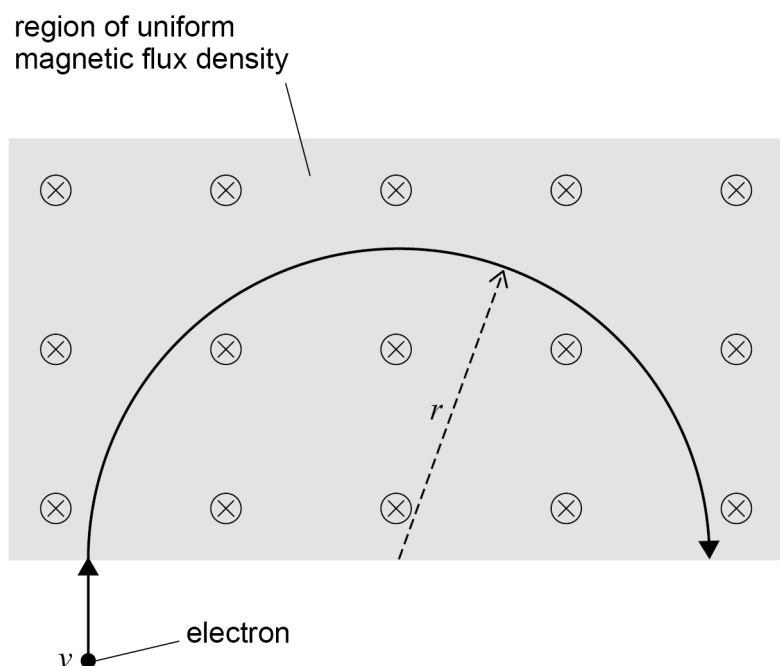


Don't waste

An electron enters a region of uniform magnetic flux density  $B$  at  $90^\circ$  to the field lines.

The electron travels at a constant speed  $v$  and a constant radius  $r$ .

The time between the electron entering the field and leaving the field is  $T$ .



The electron mass is  $m$  and the electron charge is  $e$ .

What is the value of  $B$ ?

[1 mark]

**A**  $\frac{\pi m}{Te}$

**B**  $\frac{Te}{\pi m}$

**C**  $\frac{\pi r^2 m}{Te}$

**D**  $\frac{Te}{\pi r^2 m}$

|              |
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**3****0**

Which statement is true for a charged particle in a cyclotron?

**[1 mark]**

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outside the  
box

**A** Its speed is constant as it travels through a dee.

☐

**B** Its speed is constant in the gap between the dees.

☐

**C** It accelerates in the dees only.

☐

**D** It accelerates in the gap between the dees only.

☐

**3** **2** A coil has 40 turns and an area of  $0.20 \text{ m}^2$ .

The coil rotates at a constant frequency  $f$  in a uniform magnetic flux density of  $1.5 \text{ mT}$ .

The maximum emf induced in the coil is  $0.24 \text{ V}$ .

What is  $f$ ?

**[1 mark]**

**A**  $3.2 \times 10^{-3} \text{ Hz}$

☐

**B**  $2.0 \times 10^{-2} \text{ Hz}$

☐

**C**  $3.2 \text{ Hz}$

☐

**D**  $20 \text{ Hz}$

☐

Don't waste

0 4

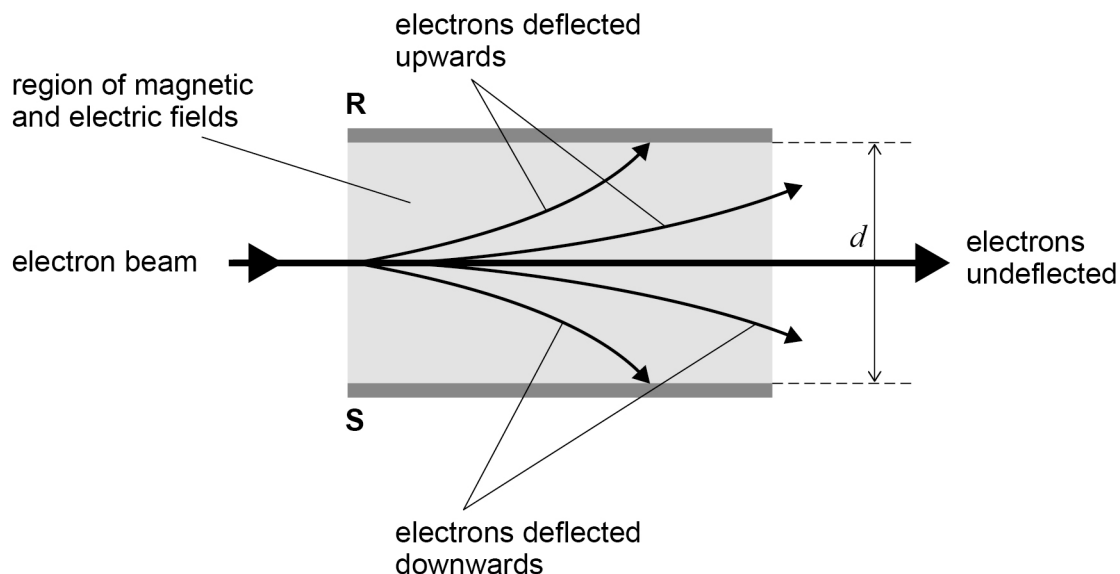
A beam contains electrons with a range of horizontal velocities. **Figure 3** shows the electron beam entering a device in a vacuum.

The device consists of two horizontal metal plates **R** and **S** separated by a distance  $d$ .

A uniform electric field of electric field strength  $E$  and a uniform magnetic field of magnetic flux density  $B$  are applied between the plates. The fields are perpendicular to each other and to the initial velocity of the electrons.

Only electrons that enter the device at a particular velocity  $v$  are undeflected. Electrons with other velocities are deflected such as in **Figure 3**.

**Figure 3**



0 4

1

The electric force on the electrons acts upwards towards plate **R**.

State the direction of  $E$ .

[1 mark]

direction of  $E$  = \_\_\_\_\_

0 4

2

State the direction of  $B$ .

[1 mark]

direction of  $B$  = \_\_\_\_\_



**0 4 . 3** The forces acting on the undeflected electrons are balanced.

Show that

$$v = \frac{E}{B}$$

**[1 mark]**

**0 4 . 4** There is a potential difference of 410 V between the metal plates.

$$d = 12 \text{ cm}$$

$$B = 0.35 \text{ T}$$

Calculate  $v$ .

**[2 marks]**

$$v = \text{_____} \text{ m s}^{-1}$$

**0 4 . 5** State and explain the direction of deflection of electrons that enter the device with a velocity greater than  $v$ .

**[3 marks]**

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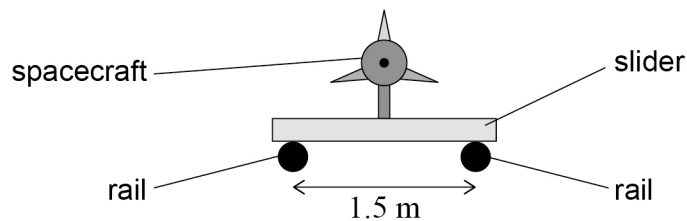
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A rail launcher is a system designed to launch a spacecraft using magnetic forces.

The launcher consists of two parallel metal rails connected to a power supply. A metal slider carries the spacecraft on the rails.

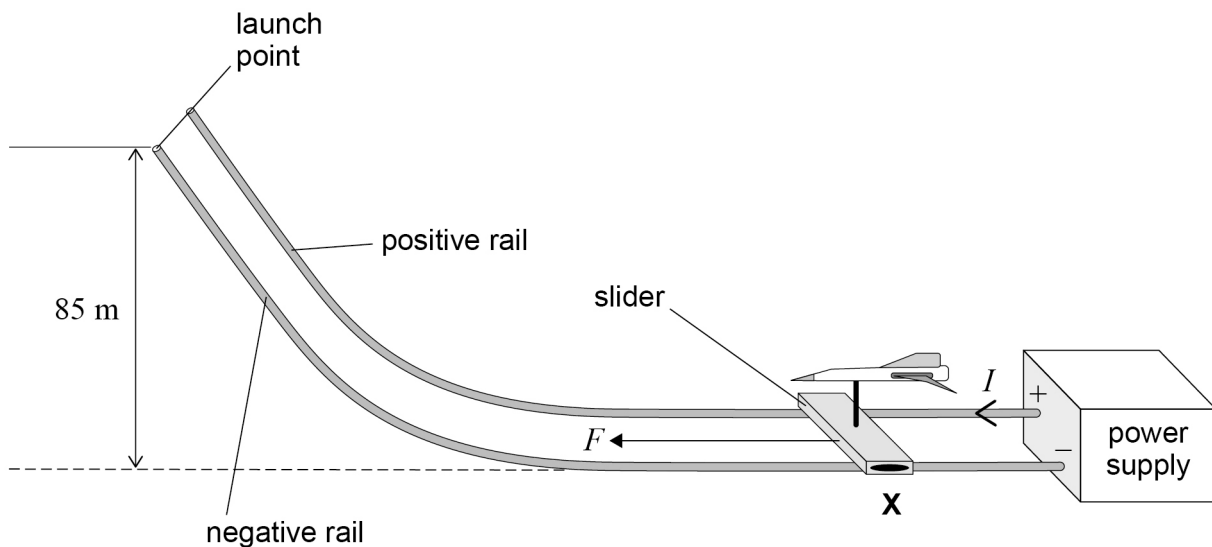
**Figure 4** shows the spacecraft and slider on the rails. The distance between the points where the slider makes contact with each rail is 1.5 m.

**Figure 4**



The slider is placed at **X** as shown in **Figure 5**.

**Figure 5**



When the power supply is switched on there is a constant current  $I$  in the rails and slider.

The current in the rails produces a magnetic field that passes through the slider. This magnetic field can be assumed to be vertical in the region of the slider. A force  $F$  acts on the slider.



0 3 . 1

State and explain the direction of the magnetic field in the slider.  
Refer to the direction of  $F$  in your answer.

[1 mark]

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In one test of the system, the slider and spacecraft are initially at rest at **X** in **Figure 5**.  
The power supply is switched on and the magnetic field that passes through the slider  
has an average flux density of 2.5 T.  
The magnitude of  $F$  is  $1.8 \times 10^7$  N.

0 3 . 2

Calculate  $I$ .

[1 mark]

$I =$  \_\_\_\_\_ A

0 3 . 3

The total mass of the spacecraft and slider is 125 kg.

The spacecraft and slider travel 250 m before reaching the launch point.  
The launch point is 85 m higher than **X**.

Assume that the magnitude of  $F$  is constant and always acts in the same direction as  
the velocity of the slider.  
Friction and drag forces are negligible.

Determine the speed of the spacecraft and slider when they reach the launch point.

[4 marks]

speed = \_\_\_\_\_  $\text{m s}^{-1}$

Question 3 continues on the next page

Turn over ►



Scientists want to increase the launch speed of the spacecraft from the value obtained in the test. The launch point remains 85 m higher than the starting position.

The gain in potential energy is negligible compared to the gain in kinetic energy. The current remains constant as the slider moves along the rails.

0 3 . 4

One scientist suggests increasing the launch speed by increasing the horizontal section of the rails so that the total length travelled by the slider is doubled.

Discuss the effect that this change would have on the launch speed.

[2 marks]

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0 3 . 5

The average flux density of the magnetic field in the region of the slider is directly proportional to the current in the rails.

A different scientist suggests increasing the launch speed by doubling the current in the rails and slider.

Discuss the effect that doubling the current would have on the launch speed.

[2 marks]

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7

What is the tesla (T) in fundamental (base) units?

[1 mark]

**A**  $\text{kg C}^{-1} \text{s}^{-1}$

☐

**B**  $\text{V s m}^{-2}$

☐

**C**  $\text{A kg s}^{-2}$

☐

**D**  $\text{kg A}^{-1} \text{s}^{-2}$

☐

A sinusoidal voltage with a peak value of 81 V is applied across a resistor. The mean power dissipated in the resistor is 48 W.

What **direct** current causes a power of 48 W to be dissipated in the resistor?

[1 mark]

**A** 0.42 A

☐

**B** 0.59 A

☐

**C** 0.84 A

☐

**D** 1.19 A

☐

Turn over ►



Use the information below to answer Question **09** and Question **10**.

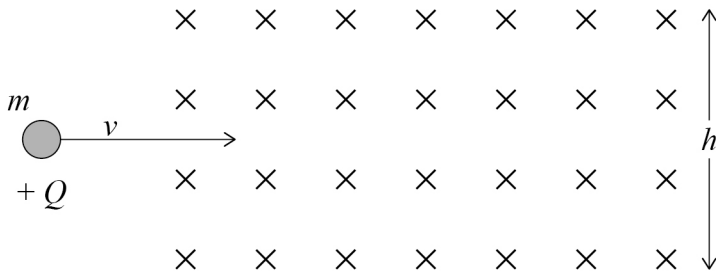
A charged particle has mass  $m$  and charge  $+Q$ .

The particle is travelling with a horizontal velocity  $v$  when it enters a uniform magnetic field of flux density  $B$ .

The magnetic field is horizontal and at right angles to  $v$ .

The region of the field has a height  $h$ .

Ignore the effect of gravity in these questions.



magnetic field is into page

0

9

What is the magnitude of the acceleration of the particle?

[1 mark]

**A**  $\frac{BQv}{m}$



**B**  $\frac{BQv}{h}$



**C**  $\frac{BQv}{mh}$



**D**  $\frac{Qv}{mh}$



Eddy currents are produced in the core of a transformer when there is an alternating current in the secondary coil.

Which statement about eddy currents is correct throughout a complete cycle of the alternating current?

[1 mark]

- A** They are eliminated using a laminated core.
- B** They reduce the flux density of the magnetic field in the core.
- C** They increase if the peak current in the secondary coil increases.
- D** They circulate around the core in the direction of the magnetic field.

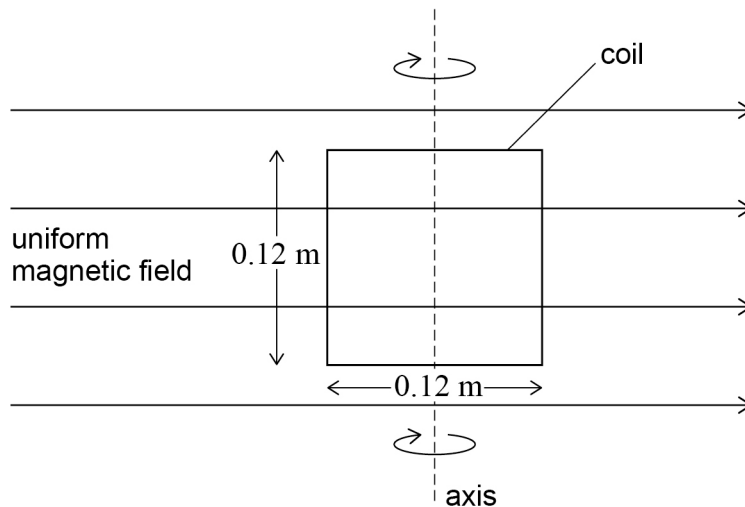
☐☐☐☐

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**1 8**

A single square coil of side  $0.12\text{ m}$  is in a uniform magnetic field that has a magnetic flux density of  $24\text{ mT}$ .

The coil is rotating at a constant angular speed about an axis at right angles to the field. The emf induced in the coil has a peak value of  $0.63\text{ V}$ .



How many times does the coil rotate in one second?

**[1 mark]**

**A** 192

☐

**B** 290

☐

**C** 552

☐

**D** 1820

☐

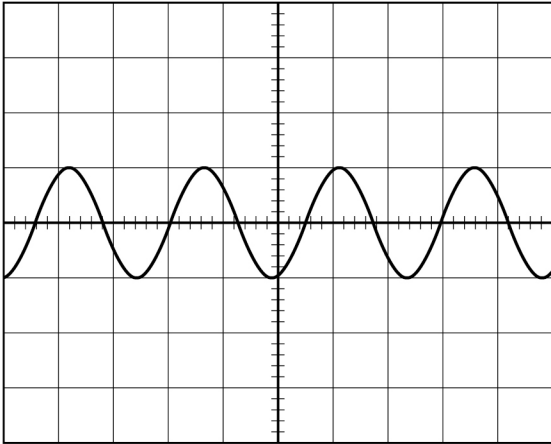
01

A student investigates the emf induced in a coil of wire when it is rotating at constant angular speed in a uniform magnetic field.

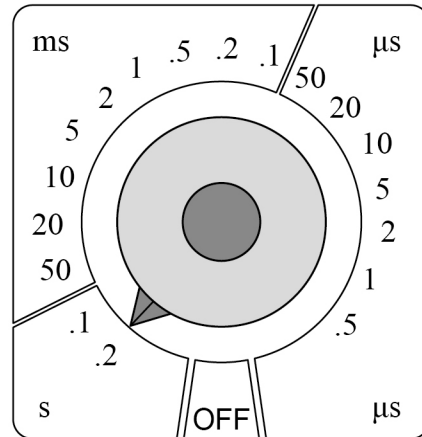
An oscilloscope is connected across the output of the coil. **Figure 1** shows the trace on the oscilloscope screen.

**Figure 2** shows the time-base setting for the oscilloscope.

**Figure 1**



**Figure 2**



time base

01

.1

Describe what the student should do to determine an accurate value for the peak emf induced in the coil.

**[3 marks]**

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0 1 . 2

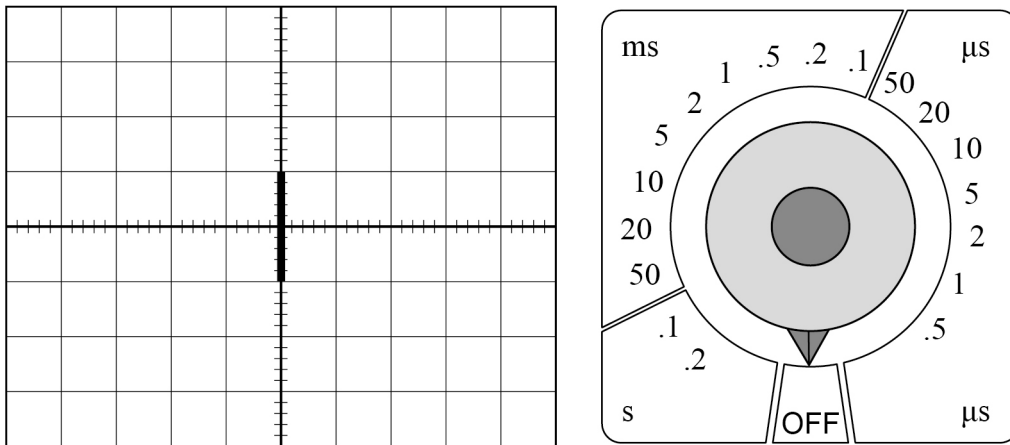
Show that the angular speed of the coil is approximately  $13 \text{ rad s}^{-1}$ .  
Use **Figure 1** and **Figure 2**.

[3 marks]

0 1 . 3

**Figure 3** shows the oscilloscope screen with the time-base set to OFF.

**Figure 3**



The student wishes to make an **accurate** measurement of the time period of the induced emf.

Calculate an appropriate time-base setting.  
Explain your answer.

[3 marks]

time-base setting = \_\_\_\_\_

\_\_\_\_\_  
\_\_\_\_\_

Question 1 continues on the next page

Turn over ►



The peak emf is  $7.5 \times 10^{-2} \text{ V}$ .

0 1 . 4

The coil has an area of  $6.0 \times 10^{-3} \text{ m}^2$  and 25 turns of wire.

Calculate the magnetic flux density of the magnetic field.

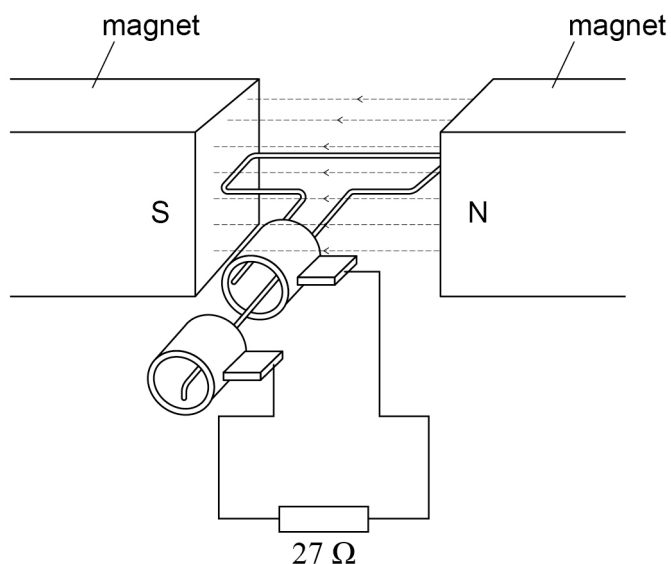
[2 marks]

magnetic flux density = \_\_\_\_\_ T

0 1 . 5

**Figure 4** shows the coil connected to a  $27 \Omega$  resistor.  
The resistance of the rest of the circuit is negligible.

**Figure 4**



Calculate the mean power dissipated in the resistor.

[3 marks]

mean power = \_\_\_\_\_ W

Don't write

0 6 . 1

Describe what is necessary for a charged particle to experience a force due to its movement in a magnetic field.

[1 mark]

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0 6 . 2

Show that the radius  $r$  of the circular path of a charged particle in a uniform magnetic field in a vacuum is given by

$$r = \frac{p}{Bq}$$

where  $p$  is the momentum of the particle,  $B$  is the magnetic flux density and  $q$  is the charge of the particle.

[2 marks]

Question 6 continues on the next page

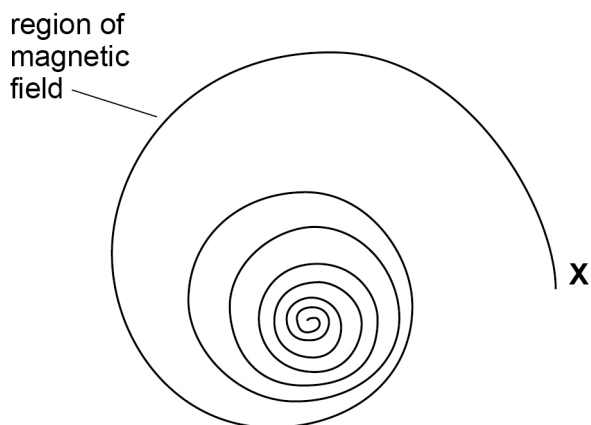
Turn over ►



A cloud chamber is a particle detector that produces visible tracks. The tracks form as charged particles ionise air molecules along their paths. Charged particles produce spiral tracks when the cloud chamber is placed in a uniform magnetic field.

**Figure 11** is a full-size image showing the track of a particle created at point **X** in a cloud chamber. The particle travels in a horizontal plane.

**Figure 11**



The magnetic field direction is into the plane of the page.

0 6 . 3

Explain why the radius of the path decreases as the particle created at **X** travels through the cloud chamber.

**[3 marks]**

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**0 6 . 4** The particle created at **X** has a charge of  $1.6 \times 10^{-19} \text{ C}$ .

The magnetic flux density in the cloud chamber is  $0.20 \text{ T}$  and acts vertically into the plane of the page.

Determine, by taking measurements from **Figure 11**, an estimate for the initial momentum of the particle.

**Figure 11** is a full-size image.

**[3 marks]**

momentum = \_\_\_\_\_  $\text{kg m s}^{-1}$

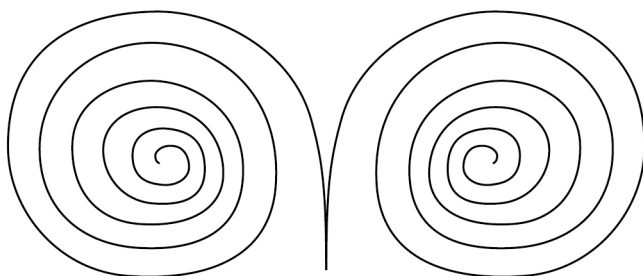
**Question 6 continues on the next page**



0 6 . 5

A gamma photon produces two particles in a pair-production event in the cloud chamber. **Figure 12** shows the tracks due to the pair-production event.

**Figure 12**



Explain what **Figure 12** indicates about the three particles involved in the pair-production event.

**[3 marks]**

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12

**END OF SECTION A**



## Section B

Each of the questions in this section is followed by four responses, **A**, **B**, **C** and **D**.

For each question select the best response.

Only **one** answer per question is allowed.

For each question, completely fill in the circle alongside the appropriate answer.

CORRECT METHOD



WRONG METHODS



If you want to change your answer you must cross out your original answer as shown.



If you wish to return to an answer previously crossed out, ring the answer you now wish to select as shown.



You may do your working in the blank space around each question but this will not be marked.  
Do **not** use additional pages for this working.

Which is a unit for magnetic flux?

[1 mark]

**A**  $\text{N m A}^{-1}$

☐

**B**  $\text{N m}^2 \text{A}^{-1}$

☐

**C**  $\text{N m}^{-1} \text{A}$

☐

**D**  $\text{N m}^{-2} \text{A}$

☐

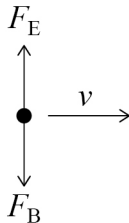
Turn over ►



Questions **18** and **19** refer to the following situation.

An electron moves to the right at a constant velocity  $v$  in a region where a magnetic field of flux density  $B$  and an electric field of strength  $E$  are applied.

Force  $F_E$  due to the electric field and force  $F_B$  due to the magnetic field act on the electron.



$F_E$  is up the page and  $F_B$  is down the page as shown.

What are the directions of  $E$  and  $B$ ?

[1 mark]

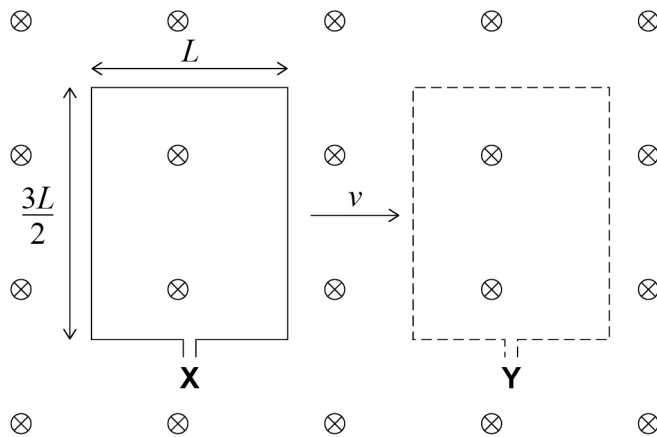
|          | Direction of $E$ | Direction of $B$ |                       |
|----------|------------------|------------------|-----------------------|
| <b>A</b> | up the page      | down the page    | <input type="radio"/> |
| <b>B</b> | up the page      | out of the page  | <input type="radio"/> |
| <b>C</b> | down the page    | into the page    | <input type="radio"/> |
| <b>D</b> | down the page    | out of the page  | <input type="radio"/> |

2 0

A rectangular coil has dimensions  $L$  and  $\frac{3L}{2}$ .

The coil moves in a magnetic field at a velocity  $v$  perpendicular to the field.

The magnetic field has a uniform magnetic flux density  $B$  perpendicular to the plane of the coil.



What is the emf induced in the coil as it moves from position X to position Y?

[1 mark]

**A**  $BLv$

☐

**B**  $\frac{3}{2}BLv$

☐

**C**  $3BLv$

☐

**D** 0

☐

Turn over for the next question

Turn over ►



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|--------------|--------------|
| Do not write | Do not write |
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0 6

The solar wind is a stream of highly energetic particles, such as protons, that flow out from the Sun.

0 6 . 1

State the condition in which a charged particle moves in a magnetic field and experiences no force.

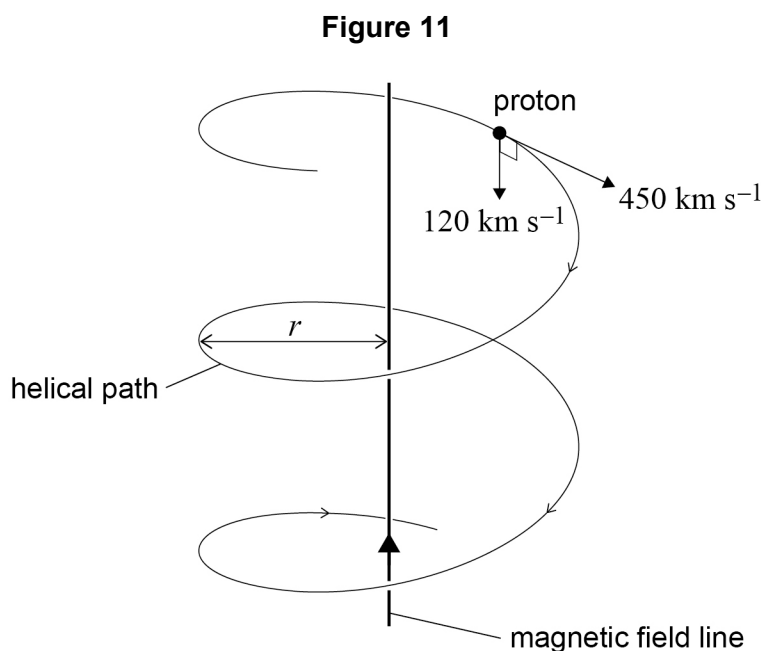
[1 mark]

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**Figure 11** shows the path a proton takes as it travels in the Earth's magnetic field. The proton moves in a helix around the magnetic field line. The radius  $r$  of the path is shown in **Figure 11**. The proton moves with constant speed.



The magnetic flux density in this region is  $1.2 \times 10^{-7} \text{ T}$ .

The velocity of the proton has two components of constant magnitude:

- component perpendicular to the magnetic field =  $450 \text{ km s}^{-1}$
- component parallel to the magnetic field =  $120 \text{ km s}^{-1}$ .



**0 6 . 2** The radius  $r$  is constant.

Calculate  $r$ .

**[3 marks]**

$r =$  \_\_\_\_\_ m

**0 6 . 3** Explain why the proton takes the path shown in **Figure 11**.

**[2 marks]**

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**Question 6 continues on the next page**

**Turn over ►**



0 6 . 4

The interaction between the solar wind and the Earth's magnetic field can cause a small change in the magnetic flux density near the Earth's surface.

Explain how this small change can cause the induction of very large voltages in power transmission lines.

Refer to Faraday's law in your answer.

**[3 marks]**

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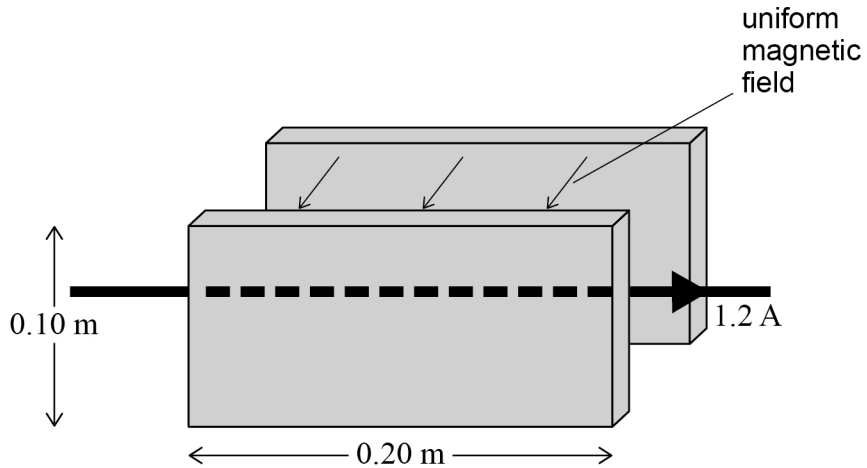
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Don't write

**1 7** Two magnets are used to create a uniform horizontal magnetic field of flux density  $0.15 \text{ T}$ . The magnetic field does not extend outside the space between the magnets.

A wire is perpendicular to the magnetic field and carries a current of  $1.2 \text{ A}$ .



What are the magnitude and the direction of the force on the current-carrying wire due to the magnetic field between the magnets?

**[1 mark]**

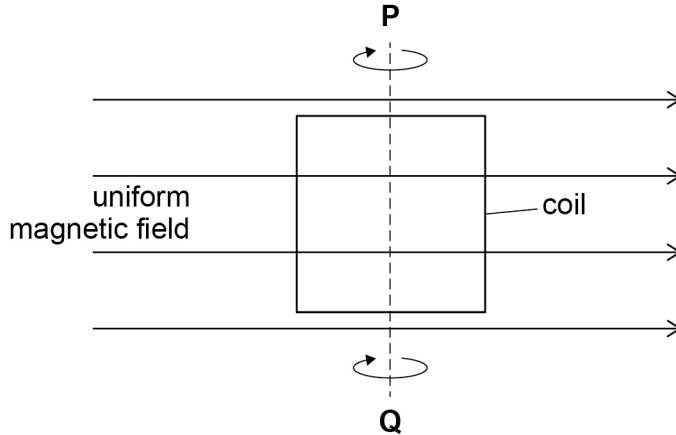
|          | Magnitude         | Direction |                       |
|----------|-------------------|-----------|-----------------------|
| <b>A</b> | $0.036 \text{ N}$ | down      | <input type="radio"/> |
| <b>B</b> | $0.018 \text{ N}$ | down      | <input type="radio"/> |
| <b>C</b> | $0.036 \text{ N}$ | up        | <input type="radio"/> |
| <b>D</b> | $0.018 \text{ N}$ | up        | <input type="radio"/> |



do not write

**1 8**

A rectangular coil has an area of  $2.5 \times 10^{-3} \text{ m}^2$  and 60 turns. The coil is in a uniform magnetic field of magnetic flux density  $0.34 \text{ T}$ . Initially, the plane of the coil is parallel to the flux lines.



The coil is then rotated through an angle of  $40^\circ$  about axis **PQ**.

What is the change in flux linkage of the coil?

[1 mark]

**A**  $0.012 \text{ Wb}$

☐

**B**  $0.018 \text{ Wb}$

☐

**C**  $0.033 \text{ Wb}$

☐

**D**  $0.039 \text{ Wb}$

☐

0 1

**Figure 1** shows a car of mass  $m$  travelling at speed  $v$  in a horizontal circle of radius  $r$ . **Figure 2** is a cross-sectional view of the track showing that it is inclined at an angle  $\theta$  to the horizontal.

Figure 1

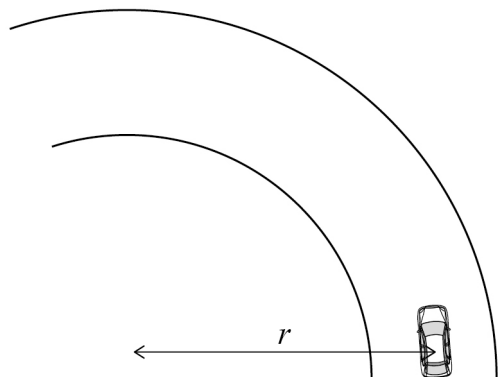
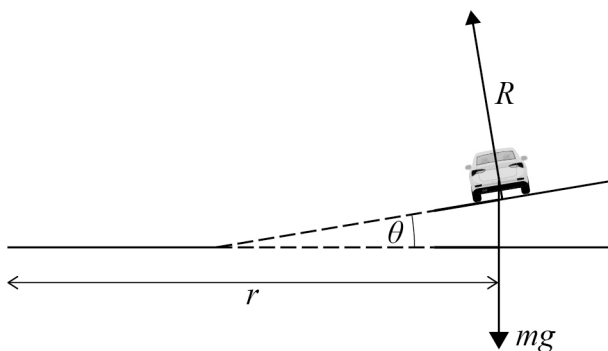


Figure 2



The sum of the normal reactions of the track acting on the wheels of the car is  $R$ . Assume that  $R$  acts through the centre of mass of the car.

0 1

. 1

The centripetal force is provided entirely by the horizontal components of  $R$ .

Show that  $v$  is given by:

$$v = \sqrt{rg \tan \theta}$$

[3 marks]



0 1 . 2

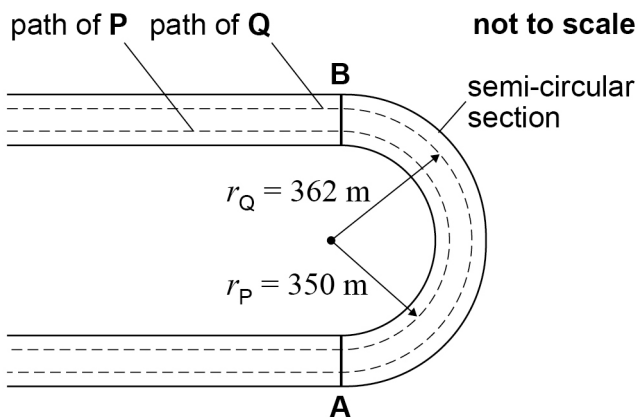
Two cars **P** and **Q** travel around another angled track.

**Figure 3** shows the plan view of the track. The section from **A** to **B** is semi-circular.

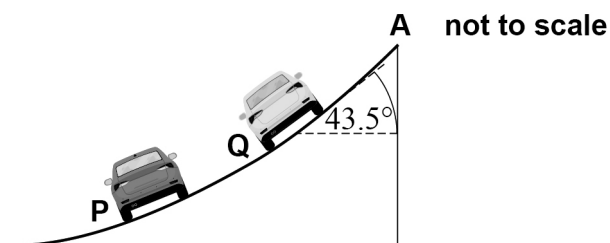
**Figure 4** shows the cross-section of the track at the point **A** where the cars enter the semi-circular section.

The steepness of the angled track increases with radius.

**Figure 3**



**Figure 4**



Both cars travel with constant speed and pass **A** at the same time.

**P** travels at a constant speed of  $54.6 \text{ m s}^{-1}$ . The radius  $r_P$  of its path is 350 m.

The radius  $r_Q$  of the path of **Q** is 362 m.

**Q** travels at speed  $u$  where  $u = \sqrt{r_Q g \tan \theta}$

$\theta$  for car **Q** is  $43.5^\circ$ .

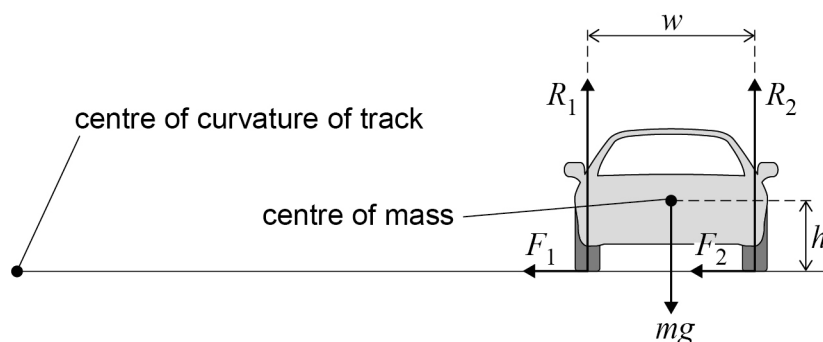
Determine whether **P** or **Q** passes **B** first.

[3 marks]

0 1 . 3

**Figure 5** shows a car travelling around a horizontal circular track. The speed of the car increases as it travels around the track.

**Figure 5**



The centripetal force is now provided by the frictional forces  $F_1$  and  $F_2$  exerted by the track on the car's wheels.

The separation of the normal reactions  $R_1$  and  $R_2$  is  $w$ .

The height of the car's centre of mass is  $h$ .

When the speed of the car is  $V$ , the left-hand wheels lift from the road surface so that  $R_1 = 0$

Show, using the principle of moments, that  $V = \sqrt{\frac{gwr}{2h}}$

**[3 marks]**

0 2 . 1

State the conditions necessary for a body to perform simple harmonic motion.

[2 marks]

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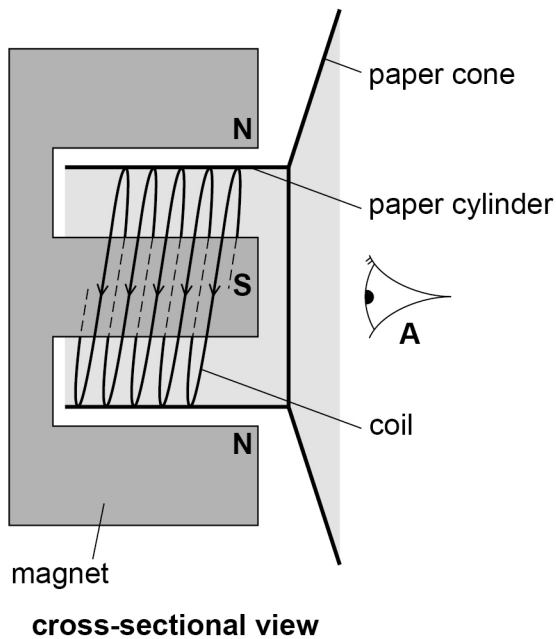
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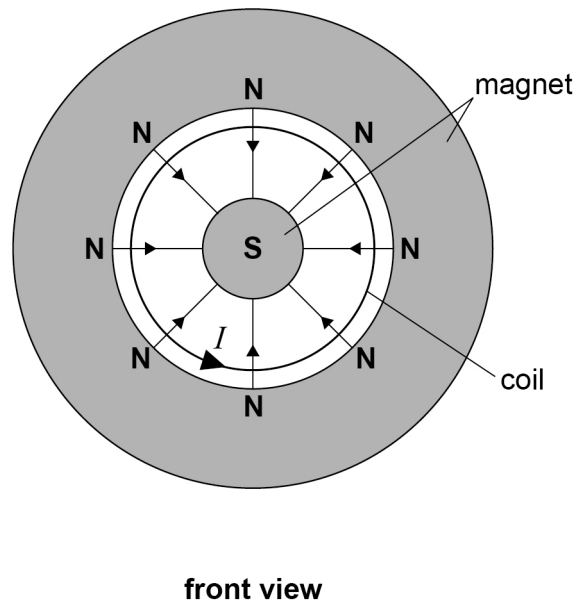
**Figure 6** shows a cross-sectional view of a loudspeaker. A coil attached to a paper cylinder and a paper cone is positioned in the cylindrical gap in the magnet.

**Figure 7** shows a front view of the coil and magnet seen from point **A**. The coil is inside the radial magnetic field created by the magnet. The current in the coil is  $I$ .

**Figure 6**



**Figure 7**



0 2 . 2

Deduce the direction of the magnetic force on the coil at the instant shown in **Figure 7**.

Explain your answer.

[2 marks]

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When a sinusoidal ac current is in the coil, the coil, cylinder and cone perform simple harmonic motion.

The coil has 40 turns and a diameter of 24 mm.

The mass of the coil, paper cylinder and cone is 7.8 g.

The peak current in the coil is 1.73 A. This produces a maximum acceleration of the coil of  $542 \text{ m s}^{-2}$ .

0 2 . 3

Calculate the magnetic flux density at the position of the coil.

[3 marks]

magnetic flux density = \_\_\_\_\_ T

The coil oscillates with a frequency of 260 Hz.

0 2 . 4

Calculate the amplitude of oscillation of the coil.

[2 marks]

amplitude = \_\_\_\_\_ m

0 2 . 5

Calculate the maximum velocity of the coil.

[1 mark]

maximum velocity = \_\_\_\_\_  $\text{m s}^{-1}$

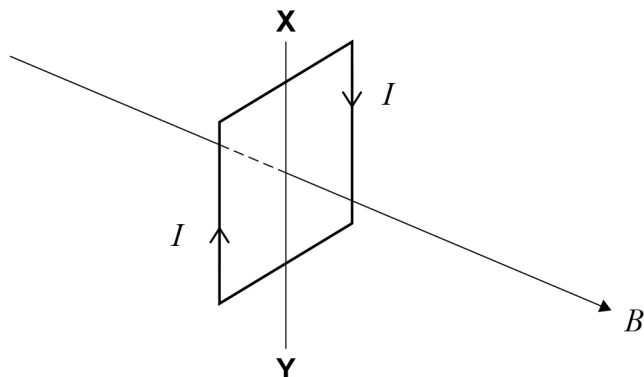


Don't write

**1 3** The diagram shows a vertical square coil in a uniform horizontal magnetic field of flux density  $B$ .

The coil can rotate about a vertical axis **XY**. The whole coil is inside the magnetic field.

The current in the coil is  $I$ .



Which statement is true?

[1 mark]

**A** No forces act on the horizontal sides of the coil.

☐

**B** The forces on each side of the coil act away from the centre of the coil.

☐

**C** The forces on the vertical sides of the coil create a turning effect.

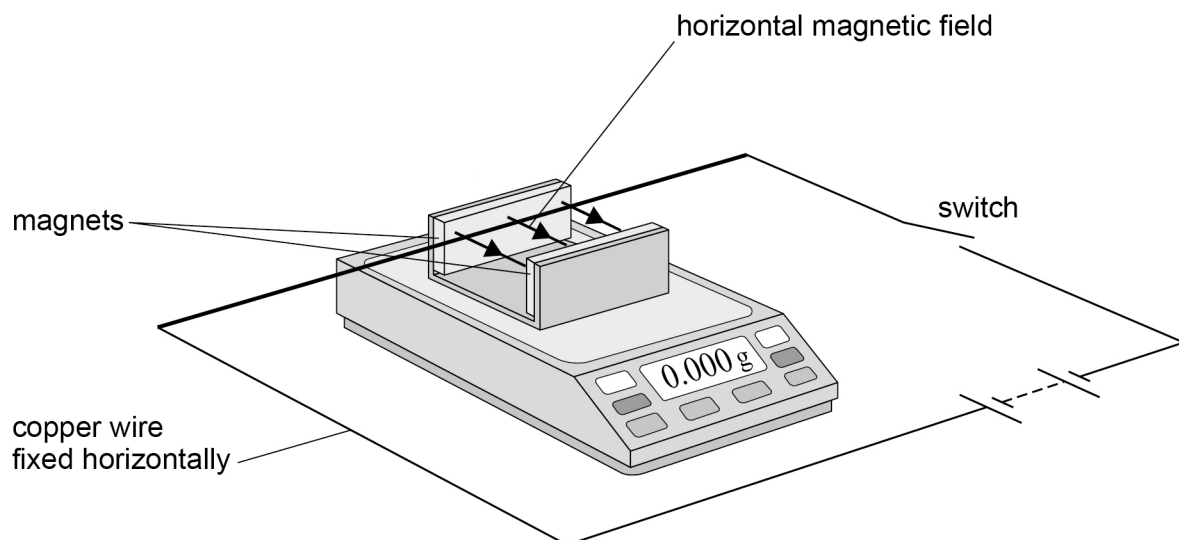
☐

**D** The forces on the horizontal sides of the coil are in opposite directions.

☐


Don't write

The diagram shows an arrangement used to investigate the force on a current-carrying conductor in a magnetic field. The copper wire is fixed horizontally in a magnetic field.



The field's magnetic flux density acts horizontally towards the front of the balance.

With the switch open the reading on the balance is set to zero.

What happens to the reading when the switch is closed?

[1 mark]

**A** It decreases before going back to zero.

☐

**B** It increases before going back to zero.

☐

**C** It decreases to a constant negative value.

☐

**D** It increases to a constant positive value.

☐

**Turn over for the next question**

**Turn over ►**

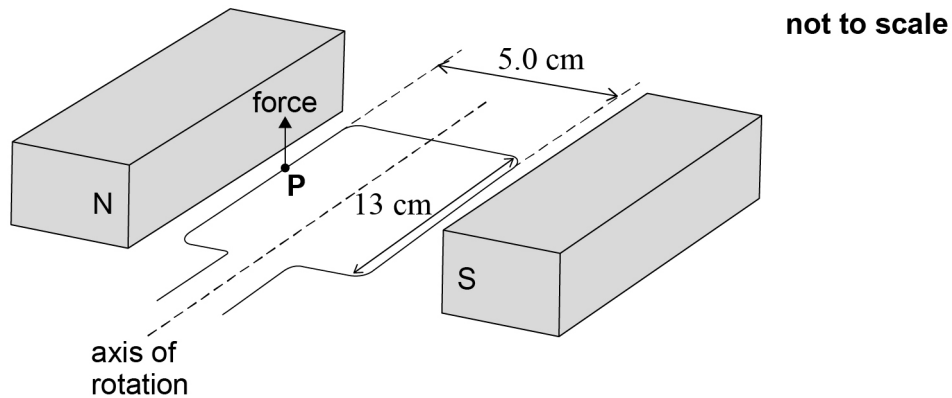


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01

**Figure 1** shows a current-carrying coil inside a uniform magnetic field. The coil rotates about an axis that is perpendicular to the direction of the field.

**Figure 1**



The coil experiences a magnetic force at point **P** as shown in **Figure 1**.

01

.1

Annotate **Figure 1** with an arrow to show the direction of the current in the coil at **P**.

[1 mark]

The uniform magnetic field has a horizontal magnetic flux density of 75 mT.

01

.2

The coil has a length of 13 cm, a width of 5.0 cm and 66 turns.

Determine the couple acting on the coil at the instant shown in **Figure 1**.

current in the coil = 1.5 A

[3 marks]

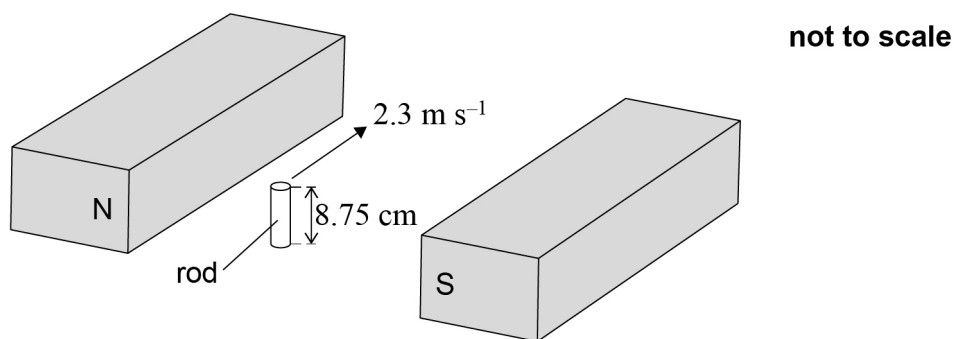
couple = \_\_\_\_\_ N m



**0 1 . 3** The coil is removed.

**Figure 2** shows a vertical metal rod of length 8.75 cm inside the magnetic field.

**Figure 2**



The rod has a constant velocity of  $2.3 \text{ m s}^{-1}$  perpendicular to the magnetic flux density.

Calculate the emf induced between the ends of the rod.

**[2 marks]**

emf = \_\_\_\_\_ V

          
**6**

**Turn over for the next question**

**Turn over ►**



Do not write

A particle with charge  $Q$  is travelling with speed  $v$  as it enters a uniform magnetic field. The field has a magnetic flux density  $B$ . The particle experiences a magnetic force that is vertically upwards as it enters the field.

Which statement must **always** be correct?

[1 mark]

A The force on the particle is  $BQv$ .

☐

B The kinetic energy of the particle is constant.

☐

C The magnetic force on the particle remains vertically upwards.

☐

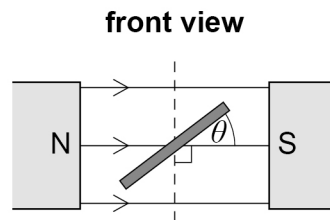
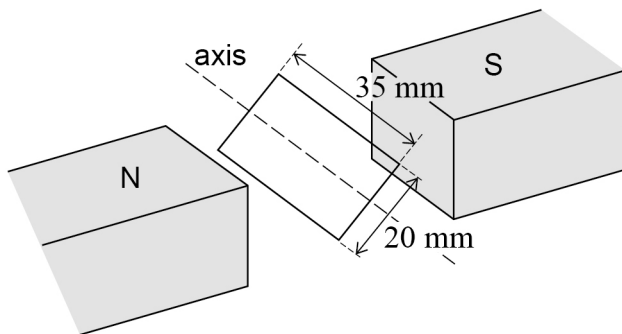
D The particle follows a circular path.

☐

Do not write

**1 7**outside the  
box

A coil with 100 turns has a length of 35 mm and a width of 20 mm. The coil is placed in a uniform magnetic field of magnetic flux density 0.40 T.



The flux linkage in the coil is  $1.0 \times 10^{-2}$  Wb when the angle is  $\theta$ .  
 $\theta$  is shown in the diagram.

What is  $\theta$ ?

[1 mark]

- A**  $1.2^\circ$  ☐
- B**  $21^\circ$  ☐
- C**  $69^\circ$  ☐
- D**  $89^\circ$  ☐

0 2 . 1

State what is meant by a magnetic flux density of 1 T.

[2 marks]

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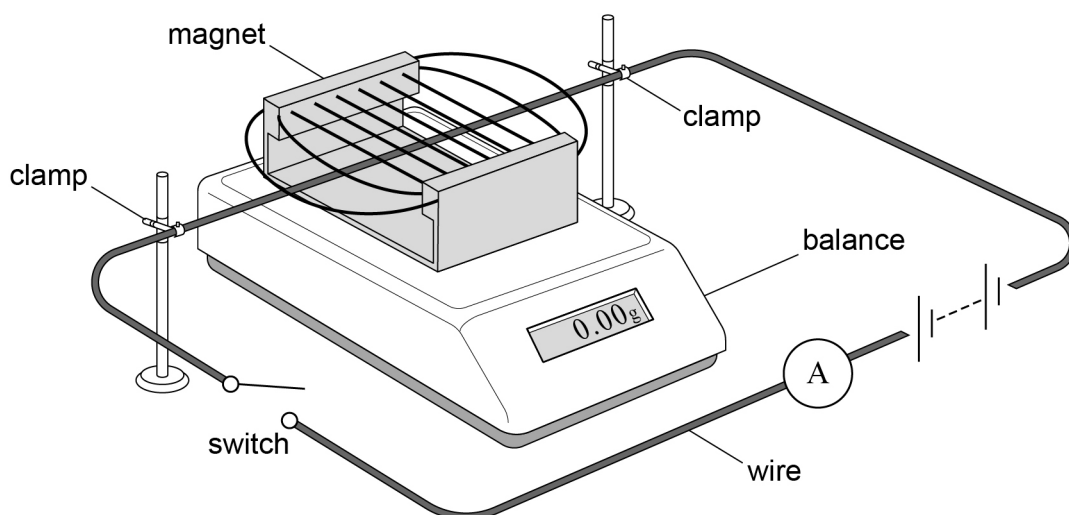
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A student does an experiment to measure  $B$ , the magnetic flux density between the poles of a magnet.

**Figure 1** shows the apparatus. Field lines for the magnetic field are shown, but without their direction.

**Figure 1**



A magnet is placed on top of a balance.

A horizontal metal wire is clamped between the poles of the magnet.

The wire is perpendicular to the magnetic field.

The wire is connected to a switch, an ammeter and a battery.

The switch is open and the balance is set to zero.



The switch is now closed and the wire remains stationary.  
The ammeter reads 2.0 A and the balance reads +0.45 g.

0 2 . 2

Draw an arrow on one of the magnetic field lines in **Figure 1** to show the direction of the magnetic field between the poles.

Go on to explain your answer.  
You should refer to appropriate physical laws.

[3 marks]

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0 2 . 3

The length of the wire between the poles of the magnet is 5.0 cm.

Calculate  $B$ .

[2 marks]

$B =$  \_\_\_\_\_ T

Question 2 continues on the next page

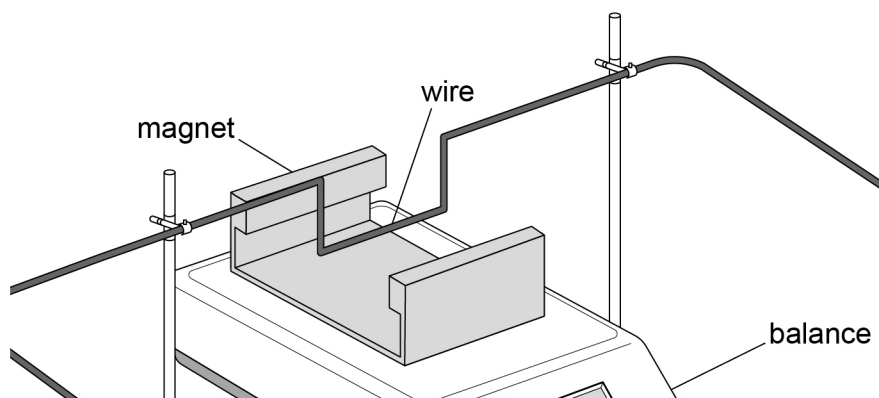
Turn over ►



0 2 . 4

Another student suggests the use of a different shaped wire, as shown in **Figure 2**.

**Figure 2**



The length of the horizontal section of the wire between the poles of the magnet shown in **Figure 2** is 5.0 cm.

Explain why the use of this wire gives a more accurate value for  $B$  than the arrangement shown in **Figure 1**.

[2 marks]

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**Turn over for the next question**

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outside the  
box*

**DO NOT WRITE ON THIS PAGE  
ANSWER IN THE SPACES PROVIDED**

**Turn over ►**

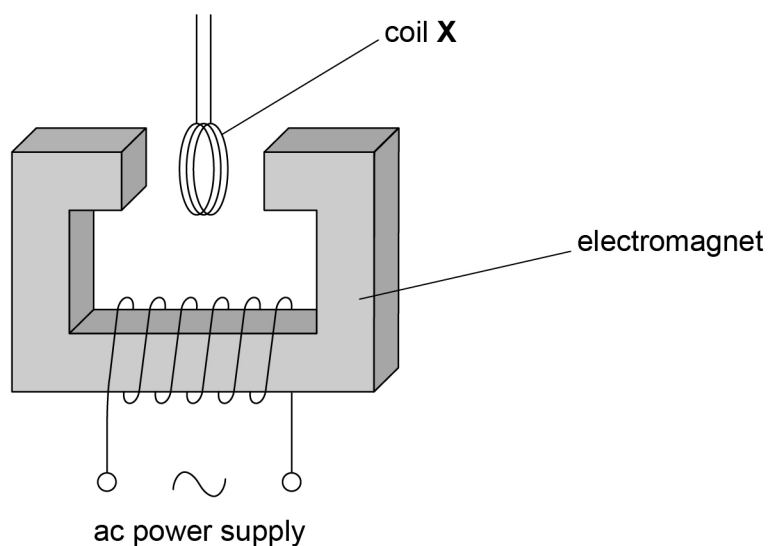


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This question is about electromagnetic induction.

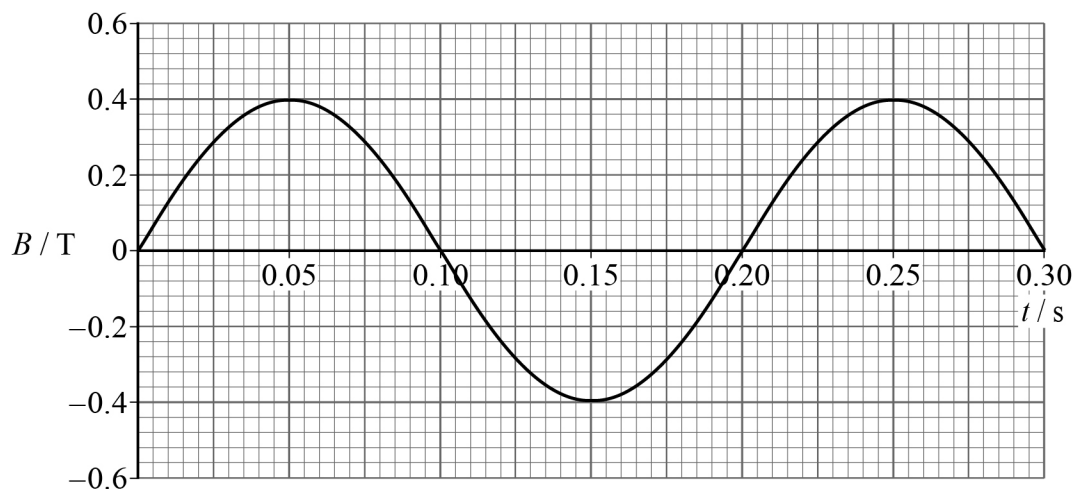
**Figure 14** shows an electromagnet connected to an ac power supply. Coil **X** is placed in the magnetic field of the electromagnet. The plane of **X** is perpendicular to the magnetic flux density  $B$  of the electromagnet.

**Figure 14**



**Figure 15** shows the variation of  $B$  with time  $t$ .

**Figure 15**



**0 8 . 1** **X** has 800 turns and an area of  $1.5 \times 10^{-4} \text{ m}^2$ .

Determine:

- the magnitude of the mean emf  $\varepsilon_1$  between  $t = 0$  and  $t = 0.10 \text{ s}$
- the magnitude of the mean emf  $\varepsilon_2$  between  $t = 0.05 \text{ s}$  and  $t = 0.15 \text{ s}$ .

**[3 marks]**

$$\varepsilon_1 = \text{_____ V}$$

$$\varepsilon_2 = \text{_____ V}$$

**0 8 . 2** The ac power supply is adjusted so that **B** has:

- a decreased peak value
- an increased period.

State and explain any changes in the frequency and amplitude of the emf induced in **X**.

**[3 marks]**

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**END OF SECTION A**



## Section B

Each of the questions in this section is followed by four responses, **A**, **B**, **C** and **D**.

For each question select the best response.

Only **one** answer per question is allowed.

For each question, completely fill in the circle alongside the appropriate answer.

CORRECT METHOD



WRONG METHODS



If you want to change your answer you must cross out your original answer as shown.



If you wish to return to an answer previously crossed out, ring the answer you now wish to select as shown.



You may do your working in the blank space around each question but this will not be marked.  
Do **not** use additional pages for this working.

1

8

A particle of mass  $m$ , charge  $Q$  and velocity  $v$  is moving in a circle in a magnetic field. The magnetic flux density  $B$  is perpendicular to  $v$ .

What is the time taken for the particle to travel once around the circle?

[1 mark]

**A**  $\frac{BQ}{2\pi m}$



**B**  $\frac{2\pi m}{BQ}$



**C**  $\frac{BQv}{2\pi m}$



**D**  $\frac{2\pi m}{BQv}$



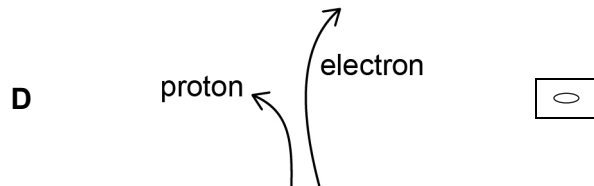
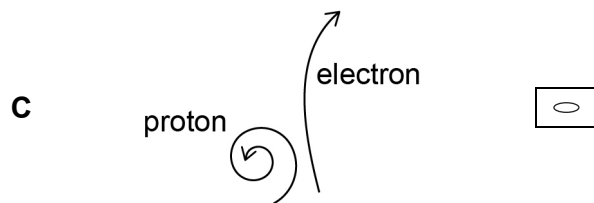
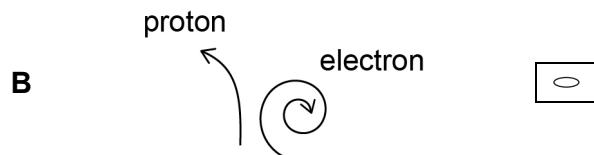
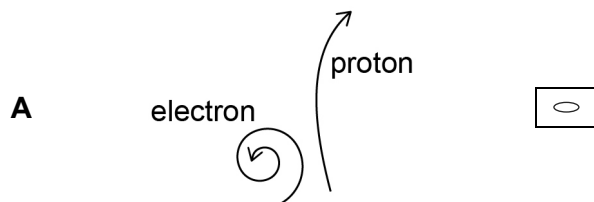
A proton and an electron travelling at the same speed enter a gas.

A magnetic field is directed into the page.

The velocities of the proton and the electron are both perpendicular to the magnetic field.

Which diagram shows the motion of the particles?

[1 mark]

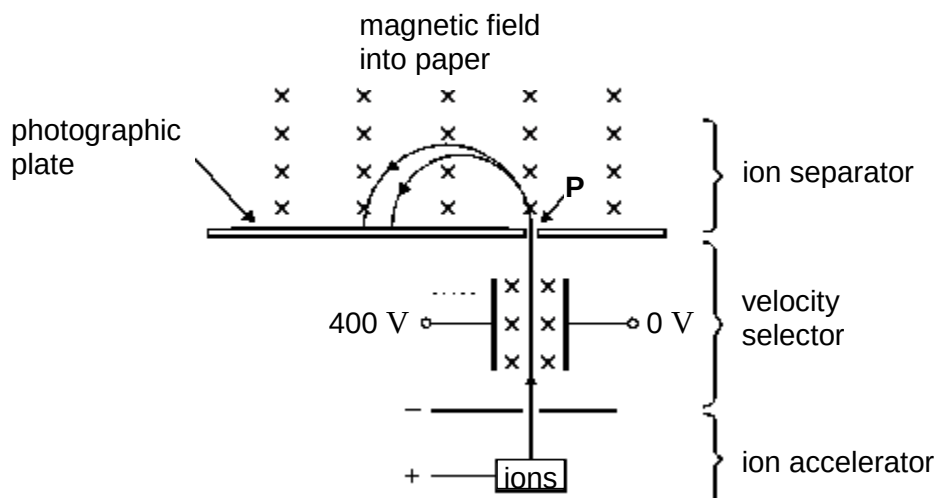


Do not write

0 4

Figure 2 shows a diagram of a mass spectrometer.

Figure 2



The magnetic field strength in the velocity selector is 0.14 T and the electric field strength is  $20\,000\text{ V m}^{-1}$ .

0 4

1

Define the unit for magnetic flux density, the tesla.

[2 marks]

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0 4

2

Show that the velocity selected is independent of the charge on an ion.

[2 marks]

0 4

3

Show that the velocity selected is about  $140\text{ km s}^{-1}$ .

[1 mark]

0 4 . 4

A sample of nickel is analysed in the spectrometer. The two most abundant isotopes of nickel are  $^{58}_{28}\text{Ni}$  and  $^{60}_{28}\text{Ni}$ . Each ion carries a single charge of  $+1.6 \times 10^{-19} \text{ C}$ .

mass of a proton or neutron =  $1.7 \times 10^{-27} \text{ kg}$

The  $^{58}_{28}\text{Ni}$  ion strikes the photographic plate 0.28 m from the point **P** at which the ion beam enters the ion separator.

Calculate the magnetic flux density of the field in the ion separator.

[3 marks]

magnetic flux density = \_\_\_\_\_ T

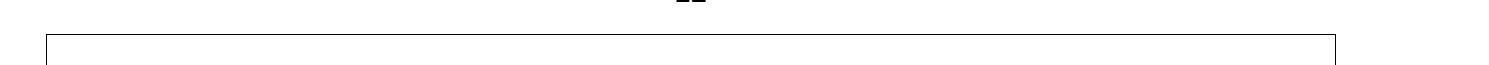
0 4 . 5

Calculate the separation of the positions where the two isotopes hit the photographic plate.

[2 marks]

separation = \_\_\_\_\_ m

**Turn over for the next question**



1 3

Charged particles, each of mass  $m$  and charge  $Q$ , travel at a constant speed in a circle of radius  $r$  in a uniform magnetic field of flux density  $B$ .

Which expression gives the frequency of rotation of a particle in the beam?

[1 mark]

**A**  $\frac{BQ}{2\pi m}$  ☐

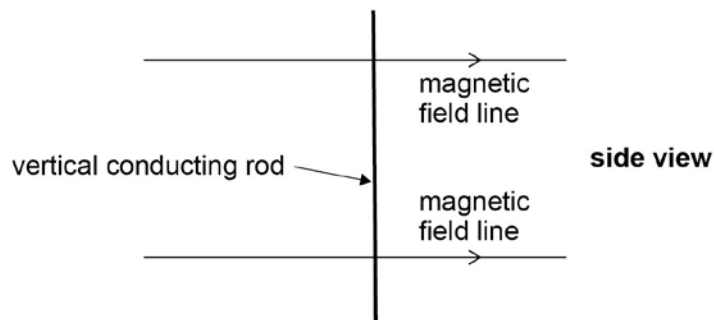
**B**  $\frac{BQ}{m}$  ☐

**C**  $\frac{BQ}{\pi m}$  ☐

**D**  $\frac{2\pi BQ}{m}$  ☐

1 4

A vertical conducting rod of length  $l$  is moved at a constant velocity  $v$  through a uniform horizontal magnetic field of flux density  $B$ .



Which of the rows gives a correct expression for the induced emf between the ends of the rod for the stated direction of the motion of the rod?

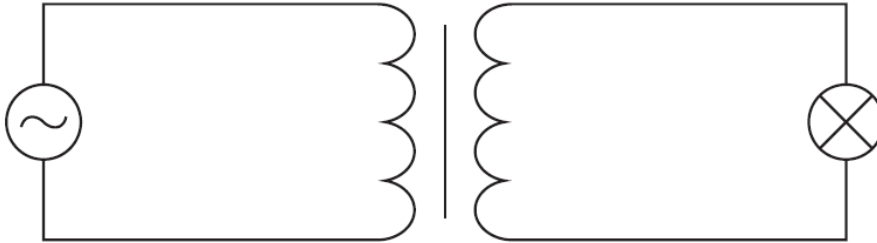
[1 mark]

|          | Direction of motion                     | Induced emf    |                          |
|----------|-----------------------------------------|----------------|--------------------------|
| <b>A</b> | Vertical                                | $\frac{B}{lv}$ | <input type="checkbox"/> |
| <b>B</b> | Horizontal at right angles to the field | $Blv$          | <input type="checkbox"/> |
| <b>C</b> | Vertical                                | $Blv$          | <input type="checkbox"/> |
| <b>D</b> | Horizontal at right angles to the field | $\frac{B}{lv}$ | <input type="checkbox"/> |

|   |   |
|---|---|
| 1 | 5 |
|---|---|

A transformer with 3000 turns in its primary coil is used to change an alternating pd from an rms value of 240 V to an rms value of 12 V.

When a 60 W, 12 V lamp is connected to the secondary coil, the lamp lights at normal brightness and a rms current of 0.26 A passes through the primary coil.



Which line, **A** to **D**, in the table gives correct values for the number of turns on the secondary coil and for the transformer efficiency?

[1 mark]

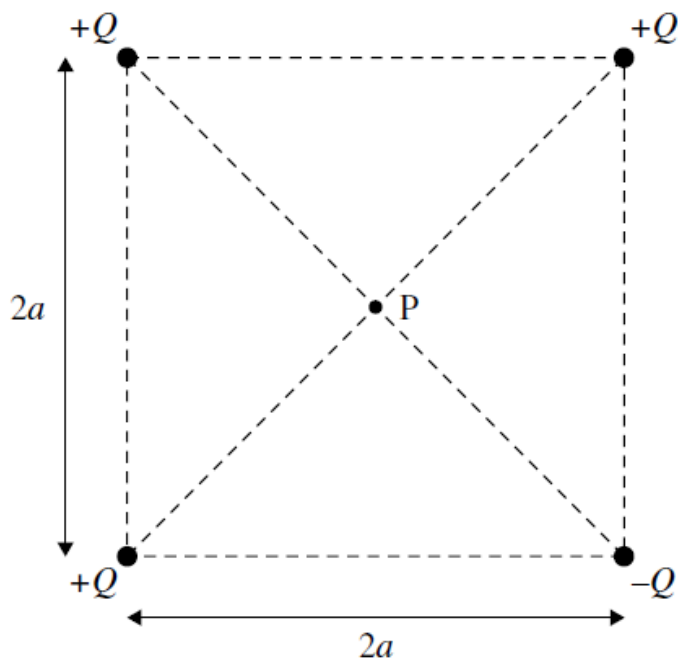
|          | number of turns on the secondary coil | efficiency |                       |
|----------|---------------------------------------|------------|-----------------------|
| <b>A</b> | 150                                   | 96%        | <input type="radio"/> |
| <b>B</b> | 60 000                                | 96%        | <input type="radio"/> |
| <b>C</b> | 150                                   | 90%        | <input type="radio"/> |
| <b>D</b> | 60 000                                | 90%        | <input type="radio"/> |

|   |   |
|---|---|
| 1 | 6 |
|---|---|

The diagram shows four point charges at the corners of a square of side  $2a$ .

What is the electric potential at P, the centre of the square?

[1 mark]



A  $\frac{Q}{2\sqrt{2}\pi a\epsilon_0}$  ☐

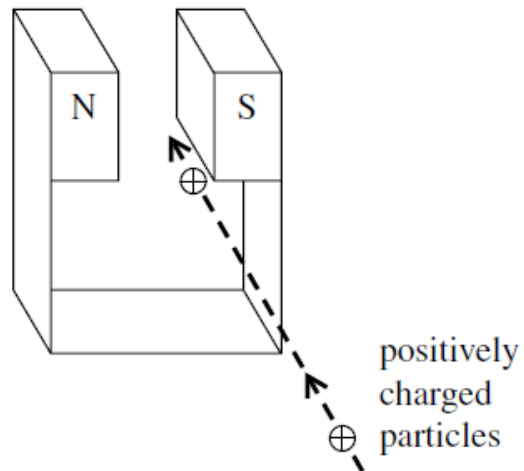
B  $\frac{Q}{\sqrt{2}\pi a\epsilon_0}$  ☐

C  $\frac{Q}{2\pi a\epsilon_0}$  ☐

D  $\frac{Q}{4\pi a\epsilon_0}$  ☐

1 7

A jet of air carrying positively charged particles is directed horizontally between the poles of a strong magnet, as shown in the diagram.



In which direction are the charged particles deflected?

[1 mark]

- A**      upwards      ☐
- B**      downwards      ☐
- C**      towards the N pole of the magnet      ☐
- D**      towards the S pole of the magnet      ☐

1 8

The half-life of a radioactive substance is 4.2 years. A freshly prepared sample of the substance contains  $5.7 \times 10^{24}$  nuclei.

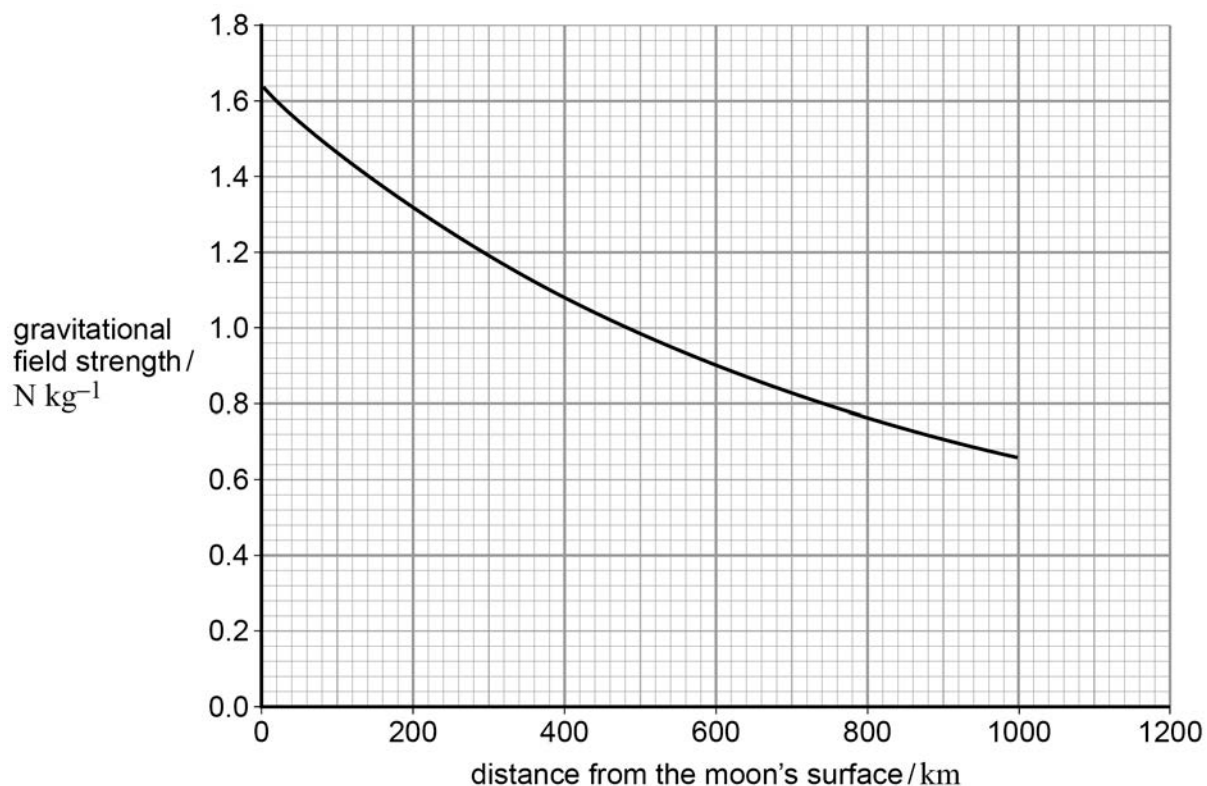
Calculate the number of nuclei of the substance remaining after 500 days.

[1 mark]

- A**       $4.5 \times 10^{24}$       ☐
- B**       $1.9 \times 10^{24}$       ☐
- C**       $4.1 \times 10^{24}$       ☐
- D**       $2.6 \times 10^{23}$       ☐

1 9

The graph shows the variation in gravitational field strength with distance in km from the moon's surface.



Estimate the energy needed to launch a 1 kg probe from the surface of the moon to a height of 400 km.

[1 mark]

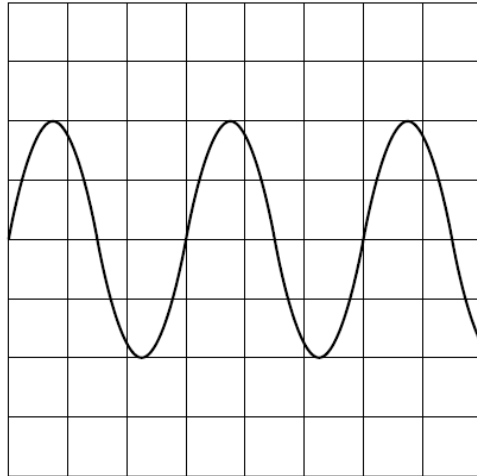
- A 67 J ☐
- B 536 000 J ☐
- C 536 J ☐
- D 136 000 J ☐

2 0

An alternating voltage supply is connected to the y-input of an oscilloscope and the following trace is obtained.

Calculate the frequency of the supply.

[1 mark]



The oscilloscope settings are: Y gain 5.0 V per division  
time base 2.0 ms per division

- A** 10 Hz ☐
- B** 3.3 Hz ☐
- C** 63 Hz ☐
- D** 170 Hz ☐

**2 1**

A square coil with 50 turns is rotated through a magnetic field 25 times a second. The magnetic flux density of the field is 45 mT. The peak emf induced in the coil is 8.0 V.

Calculate the length of the sides of the coil.

[1 mark]

- A** 2.3 cm ☐
- B** 38 cm ☐
- C** 27 cm ☐
- D** 15 cm ☐

|   |   |
|---|---|
| 2 | 2 |
|---|---|

An artificial satellite of mass  $m$  is in a stable circular orbit of radius  $r$  around a planet of mass  $M$ .  $G$  is the universal gravitational constant.

Which of the following expressions gives the speed of the satellite?

[1 mark]

**A**  $\left(\frac{GM}{r}\right)^{\frac{1}{2}}$  ☐

**B**  $\left(\frac{Gm}{r}\right)^{\frac{1}{2}}$  ☐

**C**  $\frac{Gm}{r}$  ☐

**D**  $\left(\frac{Gm}{r}\right)^{\frac{3}{2}}$  ☐